Nallabati Venkata Durga Sai Sarat Chandra

Robotics, Embedded Systems, Scientific Machine Learning & Edge AI Infrastructure
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EXECUTIVE RESEARCH & SYSTEMS BIOGRAPHY

Final-year Undergraduate Researcher in Robotics and Artificial Intelligence Engineering at Amrita Vishwa Vidyapeetham (CGPA: 9.05 / 10.00). Proven systems builder bridging low-level bare-metal firmware (STM32/ESP32 C/C++ DMA), deterministic autonomy middleware (ROS 2 Jazzy, CycloneDDS zero-copy), scientific machine learning (Physics-Informed Neural Networks, Proper Orthogonal Decomposition, SINDy nonlinear dynamics), and edge hardware acceleration (AMD Ryzen AI XDNA NPU INT8). Author of peer-reviewed autonomous vehicle research in IEEE ICRM 2025 and 25-day field campaign empirical solar desalination thermodynamic study submitted to Elsevier Desalination. National screening qualifier across ISRO YUVIKA (Top 0.1%) and NAEST experimental physics under Prof. H.C. Verma.

EDUCATION & ACADEMIC FOUNDATIONS

Amrita Vishwa Vidyapeetham — Amritapuri, Kerala, India
Bachelor of Technology (B.Tech) in Robotics and Artificial Intelligence Engineering — CGPA: 9.05 / 10.00
Duration: August 2022 – May 2026 (Expected Graduation)
Relevant Core Coursework:

- Mathematical Foundations: Multivariable Calculus, Linear Algebra & Matrix Decomposition, Ordinary & Partial Differential Equations (ODEs/PDEs), Numerical Analysis, Complex Analysis, Probability & Applied Statistics.
- Robotics & Control Systems: Spatial Kinematics & Dynamics (Forward/Inverse Kinematics, DLS Jacobian), Classical PID & Modern State-Space Control, Motion Planning & Trajectory Optimization, Kalman Filtering (EKF/UKF), LiDAR SLAM.
- Computing & Embedded Systems: Data Structures & Advanced Algorithms, Microcontrollers & Embedded Systems Architecture, Real-Time Operating Systems (FreeRTOS), Digital Signal Processing (DSP), Computer Vision.
- Artificial Intelligence & Machine Learning: Machine Learning Theory, Deep Learning & Neural Architecture, Reinforcement Learning, Physics-Informed Neural Networks (PINNs), Reduced-Order Modeling (ROM).

COMPREHENSIVE TECHNICAL SKILLS TAXONOMY

- Programming Languages: Modern C++ (C++17/20), Embedded C (C99/C11), Python (Expert), SQL (PostgreSQL, SQLite), MATLAB, R, Rust (Basics), Bash/Shell, LaTeX.
- Robotics & Autonomy Software: ROS 2 (Jazzy, Humble), CycloneDDS Middleware (Zero-Copy Shared Memory), OpenCV Perception, Probabilistic Hough Transforms, 5-DOF Damped Least-Squares Inverse Kinematics, Nav2 Stack, Gazebo/Rviz Simulation.
- Embedded Systems & Silicon Hardware: STM32 (ARM Cortex-M0/M3/M4/M7), ESP32 Dual-Core Xtensa, ESP8266, FreeRTOS Preemptive Task Scheduling, DMA-Driven Circular Buffers, Bare-Metal Register Programming, KiCAD PCB Design & Routing.
- Hardware Protocols & Buses: CAN Bus (2.0B / CAN-FD), UART (921,600 baud), Hardware SPI, I2C with 9-Clock Bus Lockup Auto-Recovery, ADC/DAC DMA Streams, PWM Timers, GPIO Interrupt Service Routines (ISRs).
- Scientific ML & Computational Physics: Physics-Informed Neural Networks (PINNs), Proper Orthogonal Decomposition (POD / SVD), Sparse Identification of Nonlinear Dynamics (SINDy), ANSYS

Fluent CFD, Carreau-Yasuda Rheology, PyTorch, SciPy.

- Edge AI & NPU Acceleration: AMD Ryzen AI XDNA NPU, ONNX Runtime (Vitis AI Execution Provider), INT8 Post-Training Quantization (PTQ), ROCm 6.x Compute Bridges, Raspberry Pi 5 Edge Computing.
- Data Analytics & Quantitative Systems: Pandas, NumPy, Scikit-Learn, Optuna Automated Hyperparameter Optimization, Time-Series Forecasting, Statistical Hypothesis Testing, Financial Risk Metrics, ETL Pipelines.
- IoT, Wireless & Power Electronics: Valetron A7670C 4G LTE Cat-1 Modem, GSM AT Command Finite State Machines, Dual-Mode WiFi ↔ 4G LTE Failover, 240V Opto-Isolated Inductive Relay Drivers, RC Snubber Transient Suppression.
- Laboratory & Diagnostic Instruments: Digital Storage Oscilloscopes (DSO), Logic Analyzers, Multimeters, Signal Generators, Soldering & SMD Rework, GDB Debugger, ST-Link / J-Link Debugging Probes.

MAJOR ENGINEERING PROJECTS & RESEARCH REPOSITORY

1. Evolution Edge — Autonomous Companion Mobile Robot (Robotics & Edge AI)

Technologies: ROS 2 Jazzy, CycloneDDS, C++17, 5-DOF IK, OpenCV, LiDAR + 9-DOF IMU EKF, Raspberry Pi 5
Role: Lead Architect & System Developer · Oct 2024 – Present

- Architected a modular ROS 2 Jazzy node architecture executing deterministic 50 Hz control loops over CycloneDDS zero-copy shared memory.
- Derived analytical and Damped Least-Squares (DLS) Inverse Kinematics for a 5-DOF robotic arm, avoiding singular configurations and executing trajectory planning in $<50\,\mu$s.
- Fused 2D planar LiDAR with 9-DOF IMU quaternion state estimation via an Extended Kalman Filter (EKF), achieving ± 1.8 cm positional tracking accuracy.
- Built differential-drive closed-loop motor velocity controllers with encoder feedback and S-curve acceleration profiling eliminating wheel slip.

2. Physics-Informed Neural Network (PINN) for Non-Newtonian Hemodynamics (SciML)

Technologies: PyTorch, SciPy, ANSYS Fluent Pulsatile CFD, Carreau-Yasuda Rheology, Adaptive NTK Balancing
Role: Lead Computational Physics Researcher · Sep 2024 – Present

- Formulated a Navier-Stokes forward/inverse physics-loss PINN pipeline to invert patient-specific blood rheology parameters from pulsatile CFD velocity fields.
- Implemented dynamic Neural Tangent Kernel (NTK) trace balancing to resolve multi-objective gradient pathologies across boundary, initial, and PDE residual loss terms.
- Achieved relative L_2 velocity prediction error below 2.4% compared against ground-truth transient ANSYS Fluent simulations.
- Accelerated hyperparameter optimization sweeps by 4.2× using automated Optuna Bayesian optimization integrated with multi-GPU PyTorch workers.

3. Reduced-Order Modeling (ROM) & Sparse Nonlinear Identification (SINDy)

Technologies: Proper Orthogonal Decomposition (POD/SVD), STLSQ Sparse Regression, Python, NumPy
Role: Sole Computational Developer · Jul 2024 – Sep 2024

- Constructed POD-Galerkin reduced basis capturing 99.2% of turbulent kinetic energy across the first 12 empirical modes from large-scale fluid datasets.
- Applied Sequential Thresholded Ridge Regression (STLSQ) with L_1 sparsity regularization to extract

parsimonious, interpretable nonlinear ODE governing dynamics.

- Delivered 1200× computational runtime speedup relative to full-order finite-volume CFD solvers while preserving structural stability over 15,000 time steps.

4. Valetron A7670C 4G LTE/GSM Smart High-Inductive Agricultural Telemetry (Industrial IoT)

Technologies: Embedded C++, STM32 ARM Cortex-M, FreeRTOS, Non-Blocking AT Parser, 240V Relays, KiCAD
Role: Lead Embedded Firmware Architect · Mar 2024 – Jun 2024

- Programmed bare-metal and FreeRTOS embedded C++ firmware driving a 4G LTE Cat-1 modem over DMA circular UART at 921,600 baud without CPU polling.
- Implemented dual-mode WiFi ↔ 4G LTE automatic failover with non-blocking AT command finite state machines recovering network links in <1.8s.
- Engineered high-voltage opto-isolated 240V relay switching with RC snubber networks designed to withstand 90V–480V agricultural inductive transient spikes.
- Integrated hardware independent watchdog (IWDG) supervision and offline EEPROM/Flash telemetry queues guaranteeing zero sensor data loss during outages.

5. Industrial Remote Acoustic Fault Diagnostic System (Edge DSP & Embedded AI)

Technologies: ESP32 Dual-Core, FreeRTOS, Edge DSP, MAX9814 AGC Pre-amplifier, Non-Blocking HTTP/WiFi
Role: System Developer · May 2024 – Aug 2024

- Built a continuous acoustic condition monitoring device processing 12-bit ADC microphone streams via dual-core FreeRTOS ring buffers.
- Designed time-domain peak-detection, RMS energy thresholding, and fast FFT spectral band analysis detecting rotating mechanical bearing defects in <12ms.
- Streamed real-time anomaly alerts over authenticated HTTP REST/WebSocket APIs with zero system lockup under heavy industrial electromagnetic interference.

6. Solar Parabolic Dish Coastal Desalination Real-Time Telemetry Campaign

Technologies: Empirical Thermodynamics, Precision Instrumentation, Sensor Fusion, Data Acquisition
Role: Experimental Co-Investigator · Mar 2024 – Jun 2024

- Conducted an intensive 25-day coastal field experimental campaign logging solar radiation, 116°C focal receiver temperatures, and hourly distillate volume.
- Demonstrated 99.5% Total Dissolved Solids (TDS) reduction (saline input 3,850 mg/L reduced to pure distillate 18 mg/L).
- Developed empirical regression models relating solar flux and thermal receiver heat transfer with high statistical significance (Adjusted $R^2 = 0.83$, $p < 0.05$).
- Authored complete manuscript submitted to Elsevier Desalination journal.

7. AMD Ryzen AI NPU Edge Vision Acceleration Engine (Silicon AI Systems)

Technologies: AMD Ryzen AI XDNA NPU, ONNX Runtime, Vitis AI EP, INT8 Quantization, C++20, ROCm 6.x
Role: Edge AI Infrastructure Engineer · Nov 2024 – Present

- Deployed INT8 post-training quantized vision transformers and convolutional models to AMD Ryzen AI NPU hardware via Vitis AI Execution Provider.
- Benchmarked on-device inference latency achieving sub-28.4ms per frame within a strict 8W SoC thermal power envelope.
- Architected ROCm 6.x heterogeneous compute pipelines spanning integrated GPU and NPU execution

nodes.

8. MAX30102 PPG Optical Biotelemetry & Pulse Oximeter (Biomedical DSP)

Technologies: Bare-Metal C, STM32 ARM Cortex-M, I2C DMA, IIR Filtering, SSD1306 OLED
Role: Firmware Developer · Jan 2024 – Apr 2024

- Developed bare-metal I2C DMA driver streaming raw red/infrared PPG photodiode data from MAX30102 sensor at 100 Hz sampling rates.
- Designed a 4th-order digital IIR bandpass filter (0.5 Hz – 5.0 Hz) eliminating motion baseline wander and high-frequency noise.
- Extracted real-time heart rate (BPM) and blood oxygen saturation (SpO_2) metrics within 200ms latency.

9. Distributed Multi-Node CAN Bus Vehicle Gateway Bridge (Automotive Systems)

Technologies: CAN 2.0B, CAN-FD, STM32, Raspberry Pi 5, CycloneDDS, Differential Signaling
Role: Hardware & Firmware Lead · Feb 2024 – May 2024

- Constructed a dual-node CAN 2.0B gateway handling 500 kbps message throughput with hardware mailbox filtering and zero packet collision.
- Engineered Transmit/Receive Error Counter (TEC/REC) Bus-Off recovery state machines for autonomous fault-tolerant network recovery.
- Interfaced transceiver to Raspberry Pi 5 over SPI/UART bridging low-level vehicle actuation to high-level ROS 2 navigation nodes.

10. Physics-Guided Diffusion Models for Thermodynamic Materials (Generative AI)

Technologies: PyTorch, Equivariant GNNs, IEEE IES GenAI Challenge Finalist, Thermodynamic Loss
Role: AI Research Lead · Aug 2024 – Nov 2024

- Developed physics-constrained generative diffusion models embedding thermodynamic equilibrium equations into reverse diffusion sampling.
- Awarded Top-5 Finalist out of hundreds of international teams in the IEEE Industrial Electronics Society (IES) Generation-AI Challenge 2024.
- Validated structural validity of generated molecular crystal lattices against Density Functional Theory (DFT) baselines with 96.8% fidelity.

PROFESSIONAL EXPERIENCE & LEADERSHIP

CLUB MIRAI (Robotics & Embedded Systems Club) — Amrita Vishwa Vidyapeetham
Embedded Systems and Robotics Developer & Hardware Lead — August 2023 – Present

- Architected distributed hardware-software integration across 10+ autonomous mobile robotics and IoT prototypes.
- Designed 4-layer custom PCBs in KiCAD including power distribution, motor driver H-bridges, MCU breakout, and sensor telemetry.
- Conducted technical workshops and mentored 40+ engineering students on bare-metal C on ARM Cortex-M, FreeRTOS, and ROS 2 middleware.

PEER-REVIEWED PUBLICATIONS & PREPRINTS

- N.V.D.S. Sarat Chandra et al., “Development and Real-Time Implementation of a Vision-Guided Autonomous Electric Vehicle,” in IEEE International Conference on Robotics and Mechatronics (ICRM 2025), Accepted & Presented, Jan 2025.
- N.V.D.S. Sarat Chandra et al., “Empirical Performance Modeling and 25-Day Field Evaluation of a Parabolic Solar Dish Distillation System,” Submitted to Elsevier Desalination, Under Review, 2024.

- N.V.D.S. Sarat Chandra, "Physics-Informed Neural Networks for Non-Newtonian Blood Rheology Inversion Under Pulsatile Flow," Preprint in preparation for Computational Mechanics, 2025.

HONORS, SCHOLARSHIPS & NATIONAL COMPETITIONS

- Global Finalist, IEEE IES Generation-AI Challenge (2024): Top-5 finalist internationally for physics-guided generative AI diffusion model.
- ISRO YUVIKA Space Science Fellow (2020): Selected among the top 0.1% nationwide by the Indian Space Research Organisation (ISRO).
- National Anveshika Experimental Physics Competition (NAEST 2021): Cleared 2 screening stages under Padma Shri Prof. H.C. Verma.
- State Rank 497 / 150,000+, AP POLYCET (2022): Achieved top 0.3% percentile in statewide engineering entrance examination.
- MTSO Mathematics & Science Olympiad Finalist (2021): Recognized for exceptional quantitative problem-solving and mathematical aptitude.